

Tensor computations

Border-rank deficiency forcing strict submultiplicativity at the tensor square

Difficulty: extreme **Importance:** interesting to the community**Status:** Solved

Rating rationale: Forcing a saving at exactly two copies for every projective variety goes beyond the known eventual-power mechanism and presents a general structural barrier. The relationship between degeneration and repeated decomposition matters to the tensor-complexity community.

Negative resolution recorded 2026-09-11. Matthew J. Colbrook's [complete manuscript](#), [Theorem 1 and Section 4 \(LaTeX source\)](#) constructs a smooth, irreducible, reduced, nondegenerate complex projective curve $X \subset \mathbb{P}^{11}$ and a point p with $\underline{R}_X(p) = 2$, $R_X(p) = 3$, and $R_{X \times X}(p \otimes p) = 9$. This refutes the exact universal implication below, using the specified Segre product with unrestricted complex coefficients. More generally, the construction delays strict submultiplicativity for any prescribed finite number of powers; the variety depends on that prescribed delay. It does not refute versions restricted to Segre or Veronese varieties or the known eventual-power saving. The full original source passed an [independent mathematical agent review](#). [Submission and verification record](#). This is documented AI-assisted verification, not external human peer review; no priority claim is made. The original target and former difficulty and importance ratings are retained below as historical context.

Problem statement

Let V be a finite-dimensional complex vector space and let $X \subset \mathbb{P}(V)$ be an irreducible reduced nondegenerate complex projective variety. Nondegenerate means that X spans $\mathbb{P}(V)$. For $p \in \mathbb{P}(V)$, define $R_X(p)$ as the least number of points of X whose projective linear span contains p . Define $\underline{R}_X(p)$ as the least s for which p lies in the Zariski closure of $\{q : R_X(q) \leq s\}$.

Embed $X \times X$ in $\mathbb{P}(V \otimes V)$ by the Segre map $([x], [y]) \mapsto [x \otimes y]$. Is it always true that

$$\underline{R}_X(p) < R_X(p) \implies R_{X \times X}(p \otimes p) < R_X(p)^2?$$

Here $p \otimes p$ denotes the projective point represented by the tensor square of any representative of p . The product variety is precisely the Segre image just defined. For tensor-rank examples, this is the ordinary tensor product with both sets of modes retained; it does not silently merge corresponding modes into a Kronecker product.

Why it matters

An exact tensor decomposition can require more terms than arbitrarily close decompositions. The conjecture asks whether that deficiency already yields a strict saving when two copies are decomposed together. It connects ill-behaved rank limits with the cost of repeated multilinear computation.

References and status check

1. E. Ballico, A. Bernardi, F. Gesmundo, A. Oneto, and E. Ventura, *Geometric conditions for strict submultiplicativity of rank and border rank*, [arXiv:1909.03811v2](#), Conjecture 1.1; *Annali di Matematica Pura ed Applicata* **200** (2021), 187–210.
2. A. Oneto and E. Ventura, *Ranks of tensors: geometry and applications*, [Bollettino dell'Unione Matematica Italiana](#) **18** (2025), 751–778, Conjecture 4.9.

Status check — 2026-09-10

Checked the original v2, Conjecture 1.1 and Theorem 4.1, the 2025 restatement, and targeted strict-submultiplicativity searches through 2026. Theorem 4.1 proves the implication for all binary forms, a

substantive subclass of the displayed target. The source also treats ternary cubics in Proposition 4.2 and its proof. A saving at some sufficiently large tensor power does not establish the required saving at the square. No general proof or counterexample was located; the product here retains both sets of factors.